Wired, wireless & mobile networks

Every network has to make the same bargain: speed and safety, or freedom to move. This module shows you how to weigh wired, wireless and mobile against the four things examiners care about — and how to justify a choice instead of just naming one.

What this key knowledge covers
This summary distils everything you need for **KK03 — Wired, wireless & mobile networks** in Cyber Security. Work through the explanations, then use the worked good-vs-weak answers to see exactly how marks are won and lost. Tick off the revision checklist at the end before a SAC or exam.

Study summary

One company, three networks

Meet **Clutch** — a Melbourne start-up whose app helps about **40,000 learner drivers** aged 16–18 book lessons, log their 120 supervised hours, and pay instructors. Clutch never thinks of itself as having "a network". It runs **three different kinds at once**, because each part of the business needs something different.

In the Carlton office: servers holding learner records, payment data and nightly backups. Needs maximum speed, reliability and security. It never moves.

Twenty staff move between desks, meeting rooms and the kitchen with laptops. They need to roam freely without plugging in — but still reach the servers.

The actual app, used by a learner and instructor in a moving car on a country road. It has to work *anywhere* — which rules out everything tethered to a building.

The strengths and limitations of **wired, wireless and mobile** network technologies, measured by **cost**, **data storage**, **reliability** and **security**. Translation: you must be able to compare the three connection types and *justify* which suits a given situation, using those four criteria as your yardstick.

Throughout this module Clutch's three needs — **server room**, **staff laptops** and **in-car app** — map almost perfectly onto wired, wireless and mobile. Keep them in mind; the exam loves "which would you choose, and why?"

Three ways to move data

Before comparing them, be crystal clear on what physically carries the data in each case. This is where careless answers lose marks.

Data travels as electrical signals down **copper cables** (Ethernet) or as light down **fibre-optic** cables. A physical wire runs from each device to a switch. Think of it as a private tunnel.

Data travels as **radio waves** through the air between a device and a nearby **wireless access point**. No cable to the device. Range is a room or a building. This is a **WLAN**.

Data travels as radio waves to **cell towers** run by a carrier (Telstra, Optus…) over **4G/5G**. Range is a whole country. Your device hops between towers as it moves.

Wired is a string-and-tin-can line between two rooms: private and clear, but you can't leave the spot. **Wireless** is talking out loud across the lounge: easy and free to wander, but anyone in the room can overhear and noise gets in the way. **Mobile** is phoning someone across the city: works almost anywhere, but you're paying a carrier and you're trusting their lines.

The four lenses examiners use

VCAA doesn't want "wired is good". It wants a comparison along set criteria. Memorise these four and apply *each one* in your answers. A neat memory hook: **C–D–R–S** — *“Cables Don't Run Slowly.”*

For a "compare" or "justify" question, walk the lenses one at a time. Even a short answer that says "*On reliability… On security… On cost…*" reads as deliberate and scores far better than a vague paragraph. The lenses are your scaffold.

□ Wired networks

A physical cable runs from every device to a switch. For Clutch, this is the **server room**: servers and backups bolted to a rack, wired with fibre and Ethernet. Nothing here ever needs to move, so wired is the obvious win.

- **Reliability:** the gold standard. No interference, no walls in the way, stable low latency. The cable does the same thing every time.
- **Data:** highest, most consistent bandwidth — fibre handles huge backups and many users at once without slowing.
- **Security:** hardest to attack. To tap the data an attacker needs *physical access* to the cable or a port inside the building.
- **Cost:** high upfront — cabling, ports, labour, and disruption to install or re-route walls and floors.
- **No mobility:** devices are tethered. You can't carry a wired laptop to a meeting room without a cable.
- **Inflexible:** adding or moving a device means running new cable, not just connecting.

Wired is a **private underground rail line** between two stations. Expensive to dig, impossible to move — but once it's there it's fast, predictable, and nobody on the street can jump on board.

□ Wireless networks (Wi-Fi)

Radio waves connect devices to an access point. For Clutch, this is the **staff laptops**: people roam the Carlton office and still reach the servers — no cable per desk.

- **Mobility:** move freely anywhere the signal reaches — desk, kitchen, meeting room.

- **Cost:** one access point serves many devices; no cabling to each one, cheap to add a new laptop or phone.
- **Convenience:** visitors and new devices join in seconds.
- **Reliability:** walls, distance, microwaves and other networks cause interference and drop-outs. Bandwidth is *shared* — more users, slower for everyone.
- **Data:** lower and more variable throughput than wired.
- **Security:** the signal radiates *beyond the walls*. Without strong encryption (WPA2/WPA3) it can be intercepted from the car park. Rogue/"evil-twin" access points and weak passwords are real risks.

Wireless is **talking out loud in a café**. Wonderfully free — you can move around and anyone can join the chat. But your voice carries to the next table, so if it's private you'd better speak in code (that's encryption). And if the café is packed, you struggle to be heard (that's congestion).

□ Mobile networks (4G / 5G)

Radio waves connect to a carrier's cell towers across the whole country. For Clutch, this is the **in-car app**: a learner and instructor driving a country road. No building means no Wi-Fi — mobile is the only option for something that moves that far.

- **Reach:** works almost anywhere there's coverage — the only realistic choice for a moving vehicle.
- **No infrastructure:** the carrier owns and maintains the towers; Clutch builds nothing.
- **Data & speed:** 5G can be very fast, and the device hands off between towers automatically as the car moves.
- **Cost:** ongoing and recurring — every device needs a SIM and data plan; caps and overage charges add up at scale.
- **Reliability:** coverage gaps in tunnels, basements and remote roads; congestion at peak times.
- **Security:** traffic is encrypted over the air by the carrier, but it's a *public, shared* network Clutch doesn't control — risks include SIM-swapping and fake-tower interception.

Mobile is the **national postal service**. You can send from almost anywhere and it reaches almost anywhere — but you pay per use, you're trusting someone else's whole system, and out in the bush the van sometimes can't get through.

Connection Trade-off Lab

Pick one of Clutch's three real needs, then try each connection type on it. Watch the **4-lens scorecard** change and see the *physical reality* in 3D — a solid cable, radio through walls, or a car handing off between towers. The exam skill here is reading the trade-off and *justifying* the fit.

The verdict line under the scorecard is exactly the kind of one-sentence justification examiners reward: a choice, tied to the criteria, with the trade-off named.

Head-to-head

No connection type "wins" outright — each is best *for a purpose*. That phrase, "it depends on the purpose", is the single most useful thing you can write.

"There is no single best network; the right choice depends on the purpose. For a fixed, data-heavy, security-critical server room, wired is best; for staff who must roam an office cheaply, wireless; for an app used in a moving car, mobile is the only viable option." — Now you've shown you can *weigh*, not just list.

Why security tracks how data travels

The security difference between the three comes down to one idea: **how easy is it to get between the sender and the data?** The more the signal travels through open air or shared networks, the more chances an attacker has.

This is why an unsecured café Wi-Fi is dangerous and a wired server room is not: the café broadcasts your data to the room. For Clutch, staff laptops on Wi-Fi must use **WPA3 plus a VPN**, and the in-car app must encrypt its own traffic on top of the carrier's, because Clutch can't trust a network it doesn't own.

Sending an *unencrypted* password over open Wi-Fi is like writing it on a postcard and reading it aloud in a crowded train. *Encrypting* it is sealing it in a locked box only the recipient can open — anyone can still grab the box, but the contents are useless to them.

Match the need to the network

Drag (or tap) each real situation into the connection you'd recommend. Think purpose first, then check it against the four lenses.

Flip cards

Best reliability, data capacity and security — but high install cost and zero mobility. Suits a fixed server room.

Cheap, mobile within a building, easy to add devices — but signal radiates out, so it needs WPA2/WPA3 encryption.

Works anywhere with coverage — the only option for a moving car. Recurring data cost; coverage gaps; public network.

Cost · Data storage · Reliability · Security. Apply each — never answer on speed alone.

The signal travels through open air beyond the walls, so it can be intercepted without physical entry — hence encryption is essential.

"No single best network — it depends on the purpose." Then pick one and defend it against the named criteria.

Command words

The command word tells you how much to write and what shape the answer takes. Tap to expand.

Quick quiz

Six multiple-choice questions. Pick an answer to see instant feedback.

P·E·L writing trainer

Aim to light all three meters. The keyword hints show what each move sounds like.

VCE-style exam questions

Five questions, **20 marks** total, building from *Identify* to *Evaluate*. Try each in your head or on paper first, then reveal the weak answer, the strong answer and the marking guide. Use the self-mark buttons — your total appears in the dock.

- 1 mark: identifies **mobile / cellular (4G/5G)**.
- 1 mark: a valid reason — works while moving / wide coverage / only option for a vehicle away from buildings.
- No mark for "Wi-Fi" or "wireless" alone (that's a building network).
- Reliability compared (up to 2): wired stable / no interference vs wireless affected by walls, range, congestion.
- Security compared (up to 2): wired needs physical access vs wireless radiates and needs encryption.
- Must **compare** (use vs/whereas), not list two separate descriptions, for full marks.
- 1 mark: wired data is enclosed in a cable / needs physical access.
- 1 mark: wireless data radiates through open air beyond the building.
- 1 mark: links this to easier interception / the need for encryption.
- Choice stated and defended (not just named).
- Criterion 1 applied with a reason (e.g. reliability — stable/no interference).
- Criterion 2 applied with a reason (e.g. security — physical access; or data — high bandwidth).
- Top answers note the limitations don't apply here (fixed location).
- Cost weighed both ways (recurring per-device vs existing shared AP): up to 2.
- Reliability weighed (5G coverage/congestion vs stable indoor Wi-Fi/wired): up to 2.
- Security weighed (carrier-encrypted but public/uncontrolled vs controlled office): up to 2.
- 1 mark: a clear, reasoned recommendation that follows from the analysis.
- Best answers note mobile still suits the in-car app — the right tool for that purpose.

Cheat sheet

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Common traps & misconceptions

Exam trap

⚠ Don't blur the line "Wireless" and "mobile" are *not* the same word for the same thing. **Wireless (Wi-Fi)** is short-range radio to *your own* access point inside a building. **Mobile (cellular)** is long-range radio to a *carrier's* towers across a region. The in-car app uses mobile, not Wi-Fi — there's no Wi-Fi on a country road.

Exam trap

Trap 1 · Wireless = mobile Calling the in-car app "Wi-Fi". Wi-Fi is your own short-range network in a building; **mobile** is a carrier's cellular network across a region. A moving car uses mobile.

Exam trap

Trap 2 · Listing, not comparing Writing three separate paragraphs that never touch. "Compare" means putting them **against each other** on the same lens: "Wired is more reliable than wireless because…".

Exam trap
Trap 3 · "Wired has no downsides" Wired is great but **expensive to install** and **can't move**. Naming a real limitation shows balanced judgement — markers look for it.

Exam trap
Trap 4 · Ignoring the four lenses Answering only on speed. A full answer touches **cost, data, reliability and security** — at least the ones the question names.

Exam trap
Trap 5 · "Wireless is just unsafe" Too absolute. Wireless is **less secure by default because the signal radiates**, but strong encryption (WPA3) closes most of the gap. Explain *why*, don't just label.

Exam trap
Trap 6 · No recommendation A "justify" or "recommend" question needs you to actually **pick one** and defend it against the criteria. Sitting on the fence scores low.

How to answer exam questions

Read the **command word** first — it sets how much to write. *State/Identify* wants a word or phrase; *Describe* wants features; *Explain* wants reasons and links; *Justify/Evaluate* wants a judgement backed by evidence. Always pull in the **scenario detail** rather than answering in the abstract. The examples below contrast a weak response with a strong one for the same question.

Q1. Identify

Clutch's app is used by a learner and instructor in a moving car on country roads. Identify the most appropriate network technology and give one reason.

✗ A weak answer
er response Wi-Fi, because it is wireless and works without cables. Why it loses marks: Confuses wireless with mobile, and "no cables" isn't the reason. Wi-Fi can't reach a car on a country road.

✓ A strong answer
er response Mobile (4G/5G), because it provides coverage across a region and follows the car as it moves, which Wi-Fi and wired cannot do.

How the marks are awarded

- 1 mark: identifies **mobile / cellular (4G/5G)**.
- 1 mark: a valid reason — works while moving / wide coverage / only option for a vehicle away from buildings.
- No mark for "Wi-Fi" or "wireless" alone (that's a building network).

Q2. Compare

A wired and a wireless network for Clutch's Carlton office, referring to reliability and security.

✗ A weak answer

er response Wired uses cables and is good. Wireless uses radio and lets you move around. Wired is more secure and wireless is less secure. Why it loses marks: Lists features but barely compares, and gives no reasons. "Good" is not evidence. Reliability is hardly addressed.

✓ A strong answer

er response On reliability, wired is more stable because the signal runs through a cable, free of the interference, walls and congestion that disrupt wireless. On security, wired is stronger because an attacker needs physical access to the cable, whereas a wireless signal radiates beyond the building and can be intercepted unless strongly encrypted (WPA3). Wireless's advantage is mobility, but on these two criteria wired leads.

How the marks are awarded

- Reliability compared (up to 2): wired stable / no interference vs wireless affected by walls, range, congestion.
- Security compared (up to 2): wired needs physical access vs wireless radiates and needs encryption.
- Must **compare** (use vs/whereas), not list two separate descriptions, for full marks.

Q3. Explain

Why a wireless network generally presents a greater security risk than a wired network. Refer to how the data travels.

✗ A weak answer

er response Wireless is less safe because hackers can get in more easily and it doesn't have good passwords. Why it loses marks: Vague and partly wrong — the issue isn't passwords. It never explains the actual mechanism (radiating signal).

✓ A strong answer

er response In a wired network the data travels inside a cable within a building the organisation controls, so intercepting it requires physical access to the wire. In a wireless network the data travels as radio waves through open air that spread beyond the walls, so an attacker nearby — even in the car park — can capture the signal without entering the building. This open path is why wireless must rely on strong encryption such as WPA3.

How the marks are awarded

- 1 mark: wired data is enclosed in a cable / needs physical access.
- 1 mark: wireless data radiates through open air beyond the building.
- 1 mark: links this to easier interception / the need for encryption.

Q4. Justify

Clutch keeps its servers and nightly backups in the Carlton server room. Justify the use of a wired network there, referring to at least two of the four criteria.

✗ A weak answer

er response Wired is the best so they should use it. It is fast and safe and that is why it is good for servers. Why it loses marks: Asserts without reasoning and never properly names or applies criteria. "Best" and "good" earn nothing.

✓ A strong answer

er response A wired network suits the server room on at least three criteria. On reliability, it gives a stable connection unaffected by interference, vital for systems running continuously. On data, fibre provides the highest, most consistent bandwidth to move large nightly backups quickly. On security, the data stays in cables inside a controlled room, so it can only be tapped by physical access — appropriate for sensitive learner and payment records. The usual downsides — cost and lack of mobility — don't matter because the servers never move.

How the marks are awarded

- Choice stated and defended (not just named).
- Criterion 1 applied with a reason (e.g. reliability — stable/no interference).
- Criterion 2 applied with a reason (e.g. security — physical access; or data — high bandwidth).
- Top answers note the limitations don't apply here (fixed location).